

INDUSTRIAL TRAINING — PROGRESS REPORT

Smart Paper Mill Production Monitoring Dashboard

Industrial training carried out at Satia Industries Limited

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Organisation	Satia Industries Limited	Department	Information Technology (IT)
Industry Supervisor	Mr. Surinder Mehra (IT Head)	Internship Period	25 May 2026 – 10 July 2026
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1. Project Overview & Objective

This report summarises the progress of the industrial-training project undertaken at Satia Industries Limited. The project delivers a web-based, real-time, role-based dashboard that replaces manual, paper-based production logging with a single digital platform for monitoring production output, quality (GSM, moisture, speed), machine status, downtime, maintenance and Overall Equipment Effectiveness (OEE). It is built entirely with free, open-source tools — a Python (Flask) REST API with an SQLite database, secured with bcrypt password hashing and JWT-based role access, and an HTML/CSS/JavaScript front-end using Chart.js, with live machine telemetry delivered over Server-Sent Events (SSE).

2. Work Completed to Date

The following functional modules have been implemented and are working end-to-end:

Module	Function delivered	Status
Authentication	JWT login, bcrypt hashing, role-based route protection	Completed
Admin Panel	Manage users, machines and thresholds; audit log (live API)	Completed
Production Entry	Output + GSM / moisture / speed; out-of-spec flagging (live API)	Completed
Downtime Logging	Events with reason and auto-computed duration (live API)	Completed
Maintenance	Log and track jobs by priority and status (live API)	Completed
Machine Monitoring	Status board with live temperature & efficiency over SSE	Completed
Dashboard	KPI cards (incl. avg OEE), trend & status charts, live feed	Completed (UI)*
Reports	Period / machine / shift filters, OEE by machine, CSV & PDF export	Completed (UI)*
Alerts	Threshold-driven alert panel with acknowledgement & filters	Completed (UI)*

** UI, charts and export are complete; these three screens currently read the seeded demonstration dataset and are being switched to the live REST API (see Section 5).*

3. Testing & Validation

The backend is validated by an automated test suite executed with pytest, each test using an isolated temporary database. **All 36 test cases pass.**

Test category	Cases	Result
Health & authentication	5	Pass
Role-based access control	4	Pass
Production CRUD & quality fields	6	Pass
Maintenance & downtime	5	Pass
Live-stream authentication	2	Pass
User management	5	Pass
Machine management & thresholds	8	Pass
Activity / audit log	1	Pass
Total	36	All Pass

4. Requirement Completion against the SRS

Area	Completion	Remark
Functional Requirements	~88%	Core FRs met; live-plant sensing out of scope
Non-Functional Requirements	~81%	Security, usability, portability satisfied
Features	~87%	All core features delivered
User Roles (4 implemented)	~88%	Viewer & Quality Inspector deferred

5. Work In Progress & Remaining Tasks

5.1 Live-data integration

- **Wire the Dashboard, Reports and Alerts screens to the live REST API** so that entered records feed the KPIs, report figures and alert list in real time (these screens currently use the seeded demonstration dataset).

5.2 Documentation consistency

- **Correct the demo-password hint to Admin@123** on the login screen (currently shows a placeholder value).
- **Fix the machine-count text on the login and dashboard screens from 9 to 20** (the actual plant has 20 machines).
- **Align the presentation deck's test count from 94 to 36** to match the report and the actual test suite.
- **Update the role-based user manuals** to cover the Downtime and Alerts pages, admin user/machine management, and server-side persistence.

5.3 Enhancements (optional / future)

- **For a production-scale demonstration**, migrate the database from SQLite to MySQL and deploy to a free cloud tier (Render).
- **Extend the role model to** add a dedicated Quality-Inspector module and a read-only Viewer role.

6. Challenges & Learnings

- Simulating live sensor data without plant hardware — solved cleanly with Server-Sent Events instead of polling or heavyweight sockets.
- Keeping documentation consistent with the evolving code — a review identified drift (e.g. test count and data-source wording) that is being reconciled in Section 5.
- Security details — correct JWT subject typing and enforcing role checks on the server, not only in the user interface.
- Designing under constraints — the whole stack had to run on a standard laptop and a normal browser, which shaped every technology choice.

7. Next Steps & Timeline

The remaining tasks in Section 5 are scheduled for completion by the end of the internship period (10 July 2026). Priority is given to the live-data integration and the documentation consistency fixes, followed by final testing, the completed user manuals and a deployment-ready build for free cloud hosting.

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Academic / industrial-training project. All production figures are simulated and industry-realistic; no confidential Satia Industries data is used.